

Algebra 1
Unit 1
Lesson 9 Practice

Name _____
Date _____
Period _____

1. Which of the following expressions can be combined?

- A. $2\sqrt{80} - 6\sqrt{20}$
- B. $3\sqrt{6} + 2\sqrt{12}$
- C. $5\sqrt{18} - 18\sqrt{5}$
- D. $-4\sqrt{50} + 5\sqrt{40}$

Three students simplified the expression $2\sqrt{90} - 7\sqrt{40}$ below. Their work is shown below.

Amy	Linda	Jaxson
$2\sqrt{90} - 7\sqrt{40}$ $2\sqrt{(9 \cdot 10)} - 7\sqrt{(4 \cdot 10)}$ $6\sqrt{10} - 14\sqrt{10}$ $-8\sqrt{10}$	$2\sqrt{90} - 7\sqrt{40}$ $-5\sqrt{50}$	$2\sqrt{90} - 7\sqrt{40}$ $2\sqrt{(9 \cdot 10)} - 7\sqrt{(4 \cdot 10)}$ $18\sqrt{10} - 28\sqrt{10}$ $-10\sqrt{10}$

2. Which student simplified the expression correctly?

3. Write a note to one of the students who made an error explaining their mistake.

4. Write a note to the other student who made an error explaining their mistake.

For question 5-10, find the sum or difference of the expression. If you cannot add or subtract the expression, explain why.

5. $\sqrt{18} - 4\sqrt{2}$

6. $\sqrt[3]{128} - \sqrt[3]{54}$

7. $7\sqrt{47} + 8\sqrt{147}$

8. $125^{\frac{1}{2}} + 320^{\frac{1}{3}}$

9. $\sqrt{200} - 2\sqrt{48} + 4\sqrt{98}$

10. $192^{\frac{1}{3}} + 2\left(81^{\frac{1}{3}}\right)$